



WELDING AUTOMATION

FINAL PROJECT 1

Lappeenranta–Lahti University of Technology LUT

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1 Introduction

This project is to create a welding procedure and simulation using 5 different welding types to develop an automated multi-sequence workflow using Visual Components OLP 10 software. The process is then studied using collision detection to ensure safe and interference-free operation. Statistical analysis is then conducted to perform a cycle efficiency of 90.22% of all robotic and auxiliary axes. It is important to study the statistical model as they provide valuable insights when improving the accuracy and efficiency. The project's primary goal is to become familiar with the software and understand the importance of welding methods and procedures.

2 Method

This project was carried out using a simulation based approach for robotic welding. The workflow was developed and validated using Yaskawa MA1900 robotic welding system. The project was initially started with ABB but due to technical difficulties, it was decided to switch back to MA1900. The components used in final project are:

- MA1900 Yaskawa Brand
- Movable Track
- Positioner
- Weld-piece block STL file

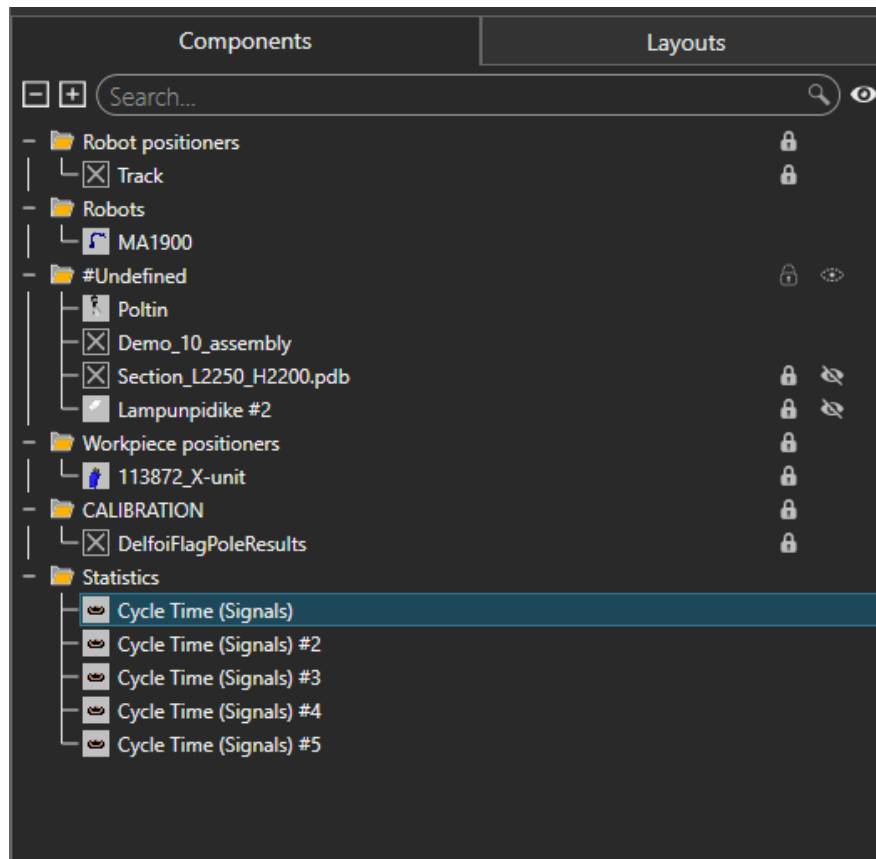


Figure 1: List of Components

It is important to consider the Weld Procedure Specification (WPS) procedure before designing the weld path based on the geometry. In robotic offline programming, these parameters must be correctly implemented in the simulation environment to ensure that the generated robot program reflects real welding conditions.

2.1 Working Environment Setup

The setup to assemble the environment was as shown in Figure 2. The workpiece is attached to the positioner plate. The torch is attached at the MA1900 robotic arm and the robotic arm is attached on top of the movable track subsequently. The robotic arm and the positioner are then linked using “Interfaces” feature which enables control options and synchronized communication between both components.

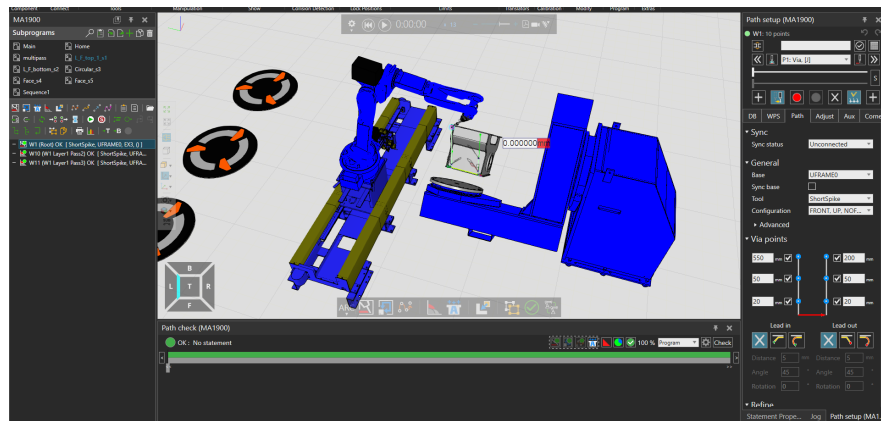


Figure 2: Work place environment

2.2 Welding Sequence Development

The welding process sequences include five different types to enable and learn programmed control welding and its feature. These sequences are then created separately in different sequence tabs then recalled into “Main” tab from which the program runs in order. It is important to create sequences in these different tabs so that it is easier to correct and track the type of welds and makes it easier for repeatability.

2.2.1 Corner Weld

The first sequence is the corner weld which was designed using 90° which enable the path to turn the corner as shown in Figure 3. 5mm offset from the corners applied in order to mitigate the weld collision and to practice the weld control corner angles.

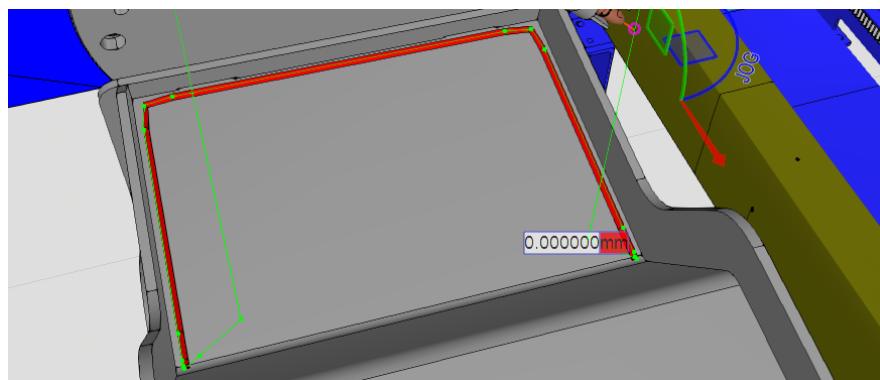


Figure 3: Corner Weld

Auto Flare Roll was used to improve torch accessibility in the tight corner. A starting flare angle of 32° was applied so that the torch will not hit the workpiece then a section option for

30mm to define the transition length for the gradual torch roll adjustment. This helped the path to maintain enough clearance and reduce collision risk.

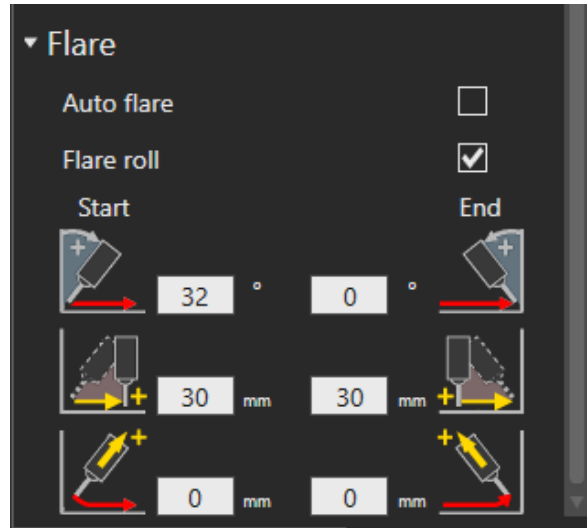


Figure 4: Auto Flare

2.2.2 Line Welds

The line passes are created linearly motion profiles. First it is important to consider the transition from the first to second weld when using large weldable block. It could collide with robotic arm which could cause disruption. In order to do so, the robotic arm was offset using both 500mm to the start and end. The weld parameters were made sure it aligned. It is also important to consider the weld torch positioning when designing the weld path to reduce the rotation inefficiency.

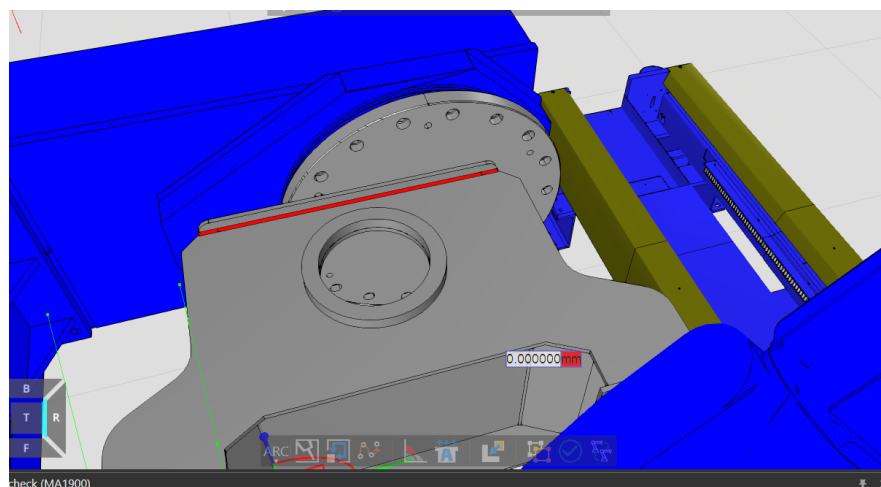


Figure 5: Line weld 1

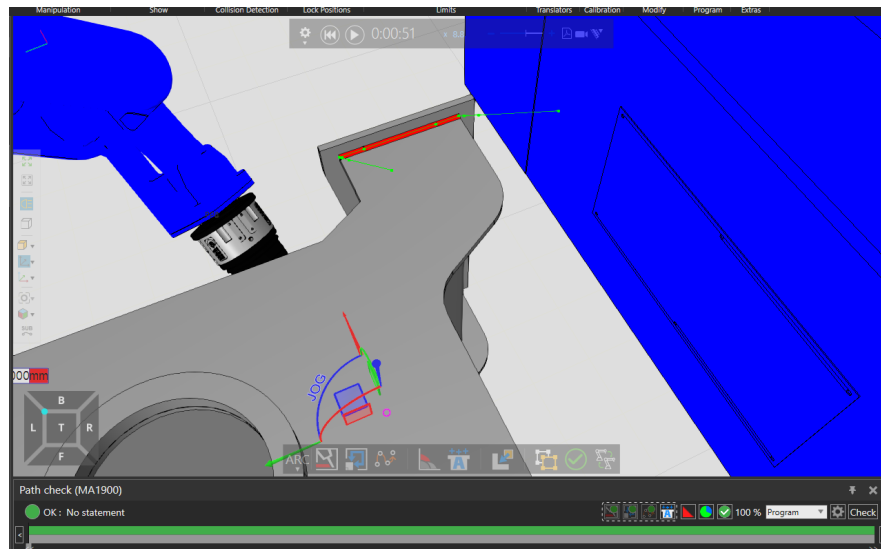


Figure 6: Line weld 2

2.2.3 Line Multi-pass Weld

Similar to the linear line weld, the initial first weld line was set up on the linear weld line. Due to the geometry that it is applied, the wire angle is not applied. 1mm offset from base and wall are applied due to the tight area even though it is not ideal for fusion quality for bead placement. The second path was reversed to achieve quicker turnaround time and provide better heat distribution.

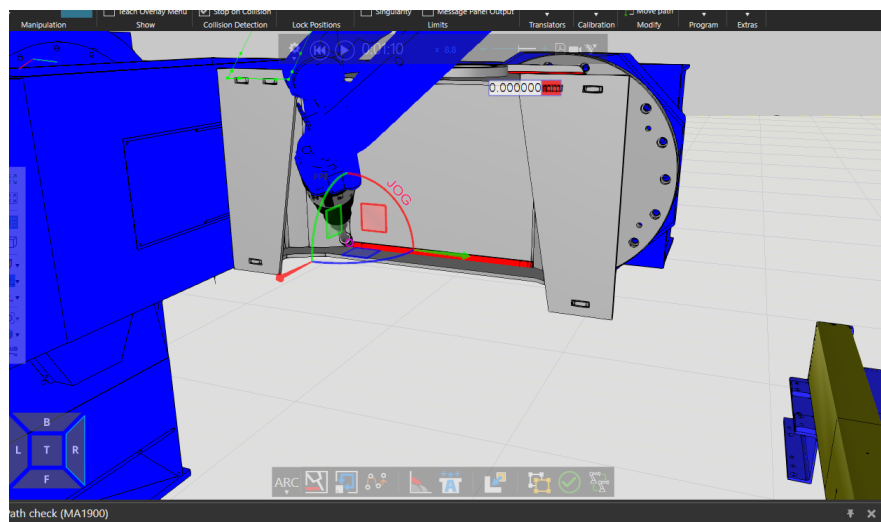


Figure 7: Multi-pass Weld

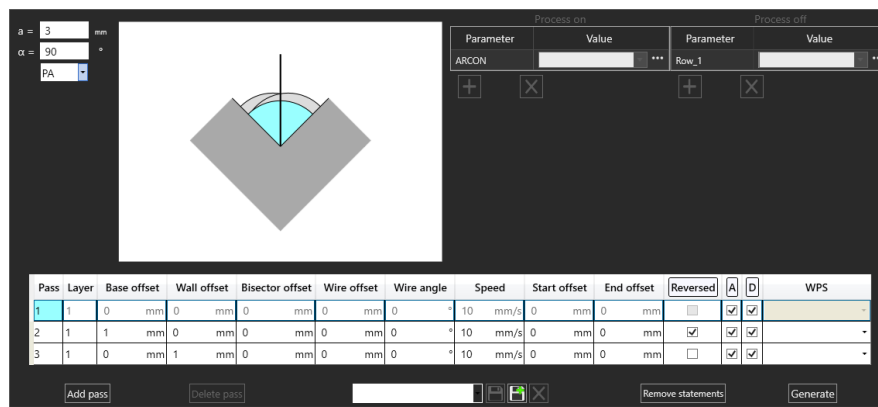


Figure 8: Multi-pass configuration

2.2.4 Circular Weld

The last sequence was for the circular weld geometry. The topology filter was edited to 360° so the path can have full circular movement as shown in Figure 17. It is important to have more points around the weld pattern of the circular path in order to have more accurate welding sequence. The positioner is then aligned my automating synchronizing with Positioner and Tracker. The A/B is selected in AUX tab and Tracker to move to align the movement. This has allowed the robotic arm and welding torch to automatically align the path.

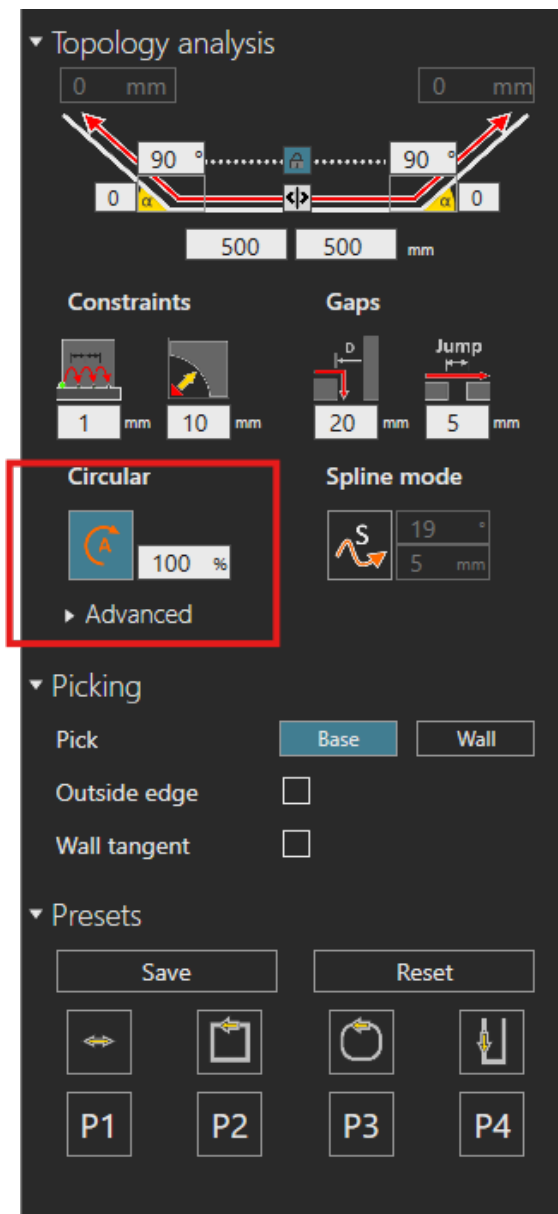


Figure 9: Topology selection for circular weld

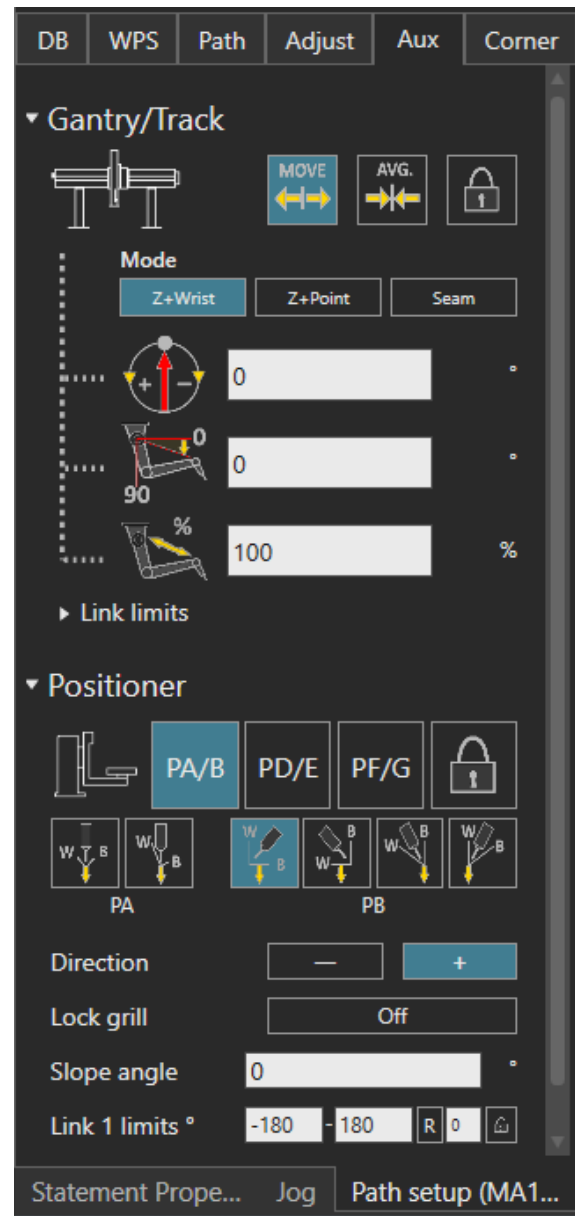


Figure 10: Synchronizing of Tracker and Positioner

The circular weld pattern can be seen in following figure. It is important to check whether the bead is directly place on the path for accurate welding.

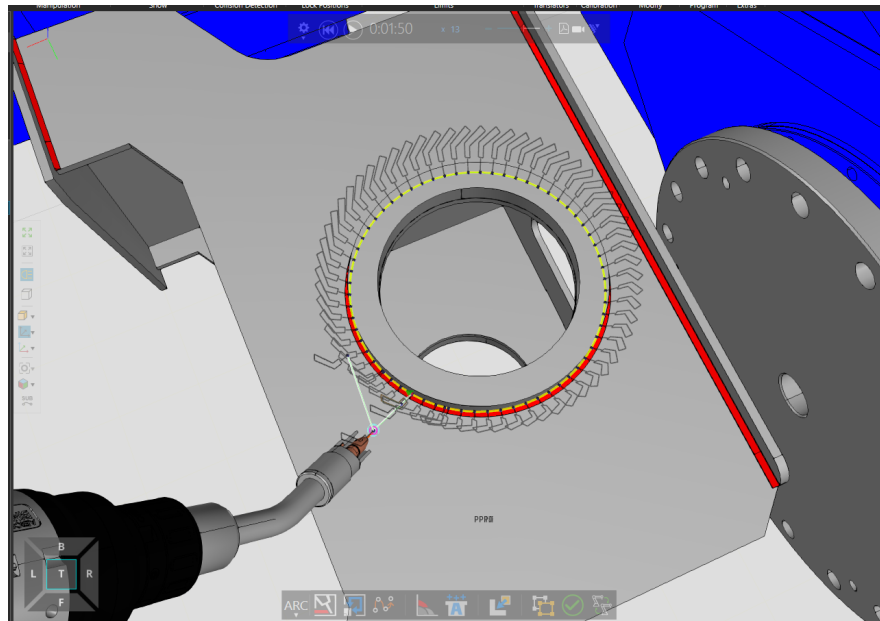


Figure 11: Selection Points for Circular Path

2.2.5 Collision Detection Setup

It is very important to validate the collision detection during welding program set up in the simulation. This not only improve the operational safety, but also helps prevent potential collisions between robot, welding torch and fixtures. Collision detection validation ensures that programmed path can be executed safely. First the collision option was selected according in Figure ?? . The Robotic arm and torch are selected in A part whereas Positioner and work-piece are selected in B part respectively. The 'Enable Detectors' and 'Stop on Collision' are also turned on to detect the possible collision during simulation.

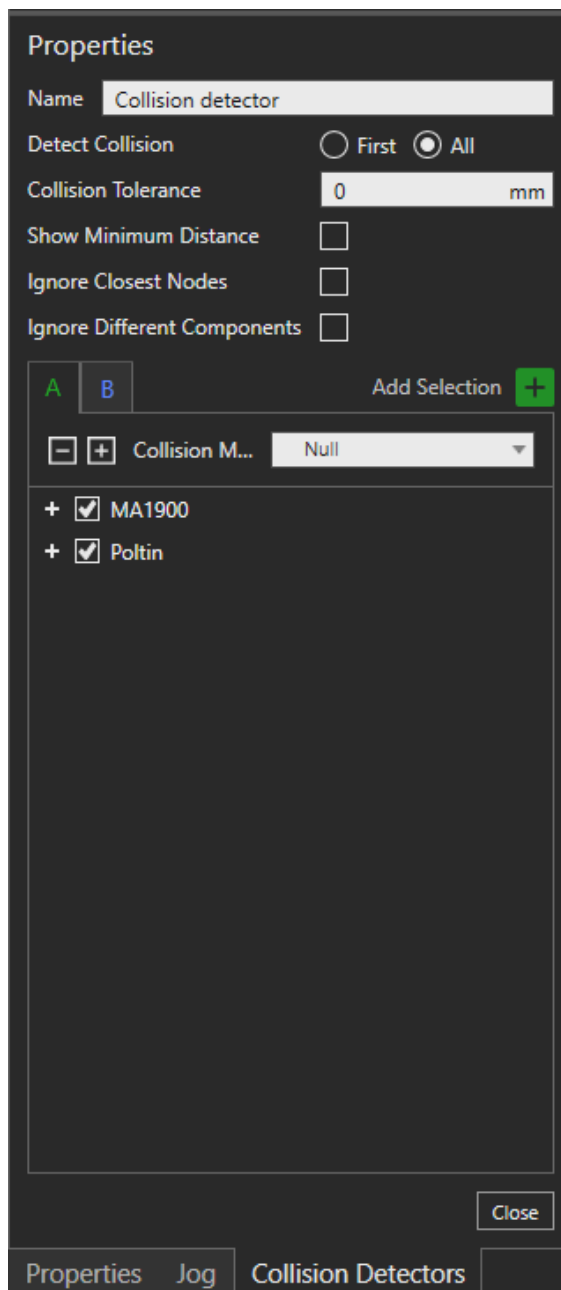


Figure 12: Collision Selection A part

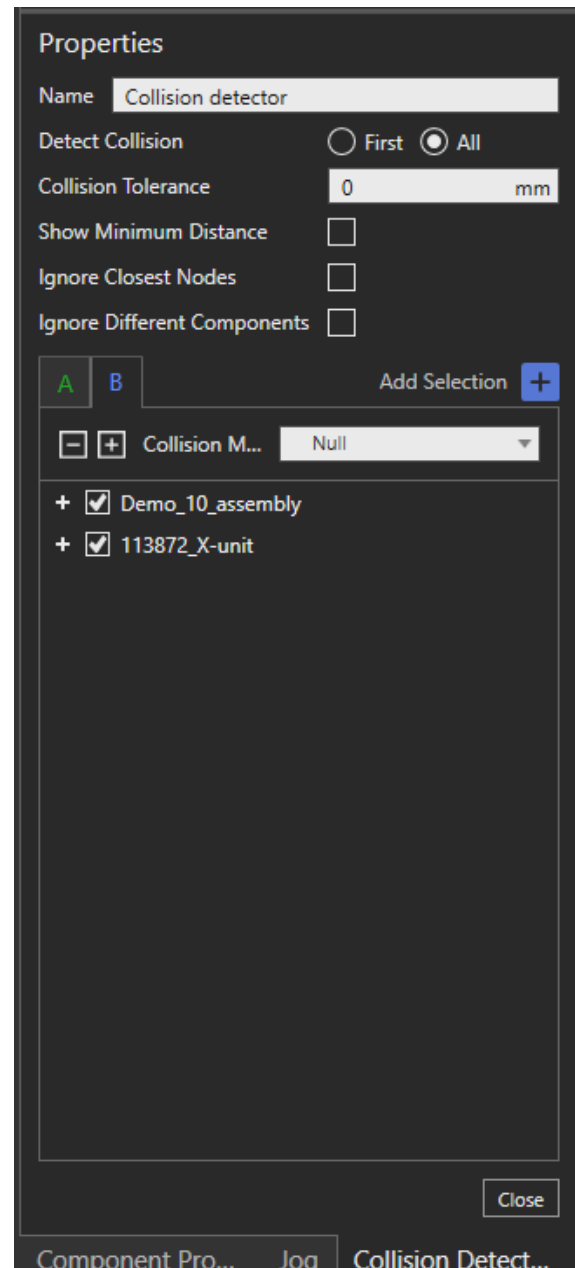


Figure 13: Collision selection B part

2.2.6 Statistical Data Collection

The statistical data collection features are also available to measure the dynamic movements of kinematic joints of robotic arm. This helps to evaluate the efficiency and performance of welding process such as cycle time, via time, joint movement and axis utilization. The collected data are then used to identify the inefficient robot motions and optimize the path planning and reduce unnecessary joint movement during the actual operation. The cycles are installed to measure 5 different weld sequences.

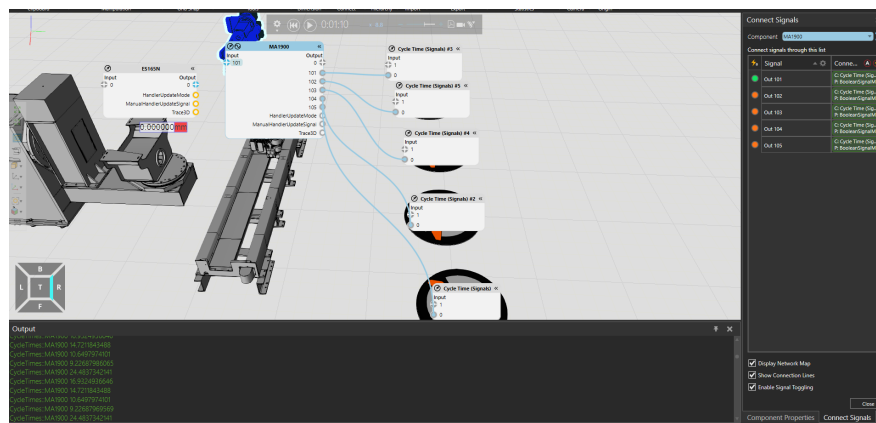


Figure 14: Installation of cycles

True and False binary output are added in the Main welding to enable data collection of each sequence. This provides an accurate breakdown of each process such as processing and idle time. Using these data, charts can be created to analyze further.

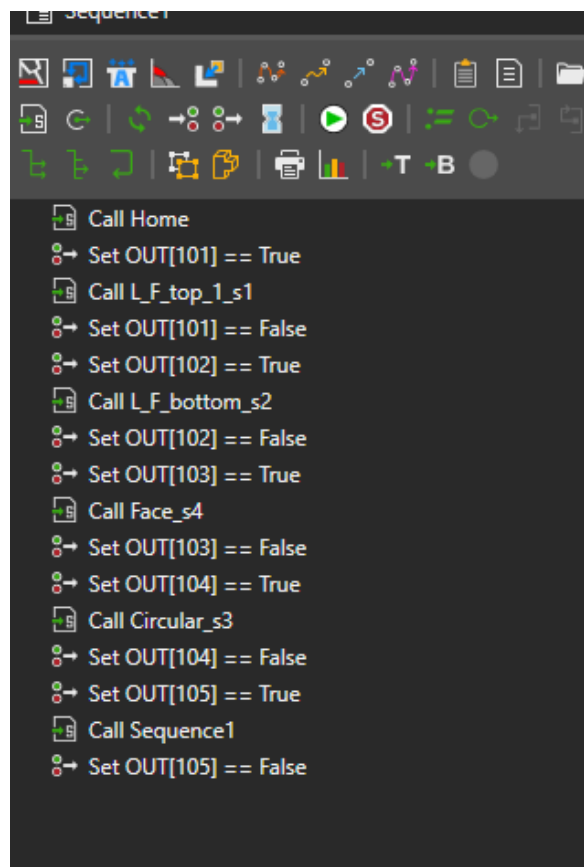


Figure 15: Binary controls after each sequence

2.2.7 Post-Processing File

The simulation data was converted to JBI files by choosing related machine type which is Yasakawa in this project. This provides post-processing file for each sequences and main file Figure 16. These files are the translated version of the simulation and they can be then transferred to robot controller.

```

/JOB
//NAME POSTPROCESSING
//POS
///NPOS 0,0,0,0,0,0
//INST
///DATE 2026/05/13 10:02
///ATTR SC,RW
///GROUP1 RB1,BS1,ST1
NOP
*POSTPROCESSING
MEMOF
CLEAR MPF
CALL JOB:HOME
DOUT OT#(101) ON
CALL JOB:L_F_TOP_1_S1
DOUT OT#(101) OFF
DOUT OT#(102) ON
CALL JOB:L_F_BOTTOM_S2
DOUT OT#(102) OFF
DOUT OT#(103) ON
CALL JOB:FACE_S4
DOUT OT#(103) OFF
DOUT OT#(104) ON
CALL JOB:CIRCULAR_S3
DOUT OT#(104) OFF
DOUT OT#(105) ON
CALL JOB:SEQUENCE1
DOUT OT#(105) OFF
END

```

Figure 16: Post-processing JBI File

3 Results

The final outcomes were developed, and the weld sequence was finalized after continuous improvements to the tool path, joint behavior and adjustments to the weld order.

3.1 Collision Detection

During the programming of each weld, collision detection was carried out to ensure that there are no collisions detected between the workpiece and the robot. When collisions were detected, tools in Path Setup were used to modify the path without hindering the integrity of the weld as presented in Figure 18. These options included refinements to flare, offsets and lead in/out. Further, simulations were run after completion of each weld to check on collisions during the workpiece/robot movement to the next weld. During which the detected collisions were removed using robot configurations and adjustments to the via points.

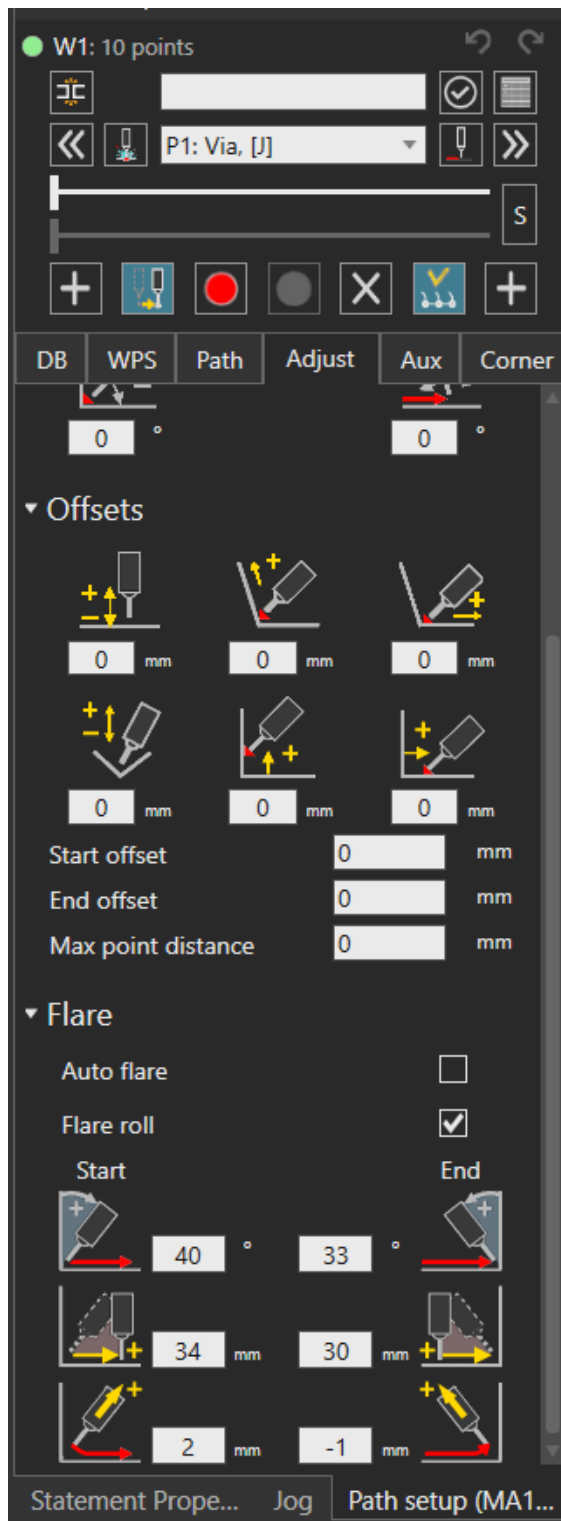


Figure 17: Configuration modifications to adjust collisions

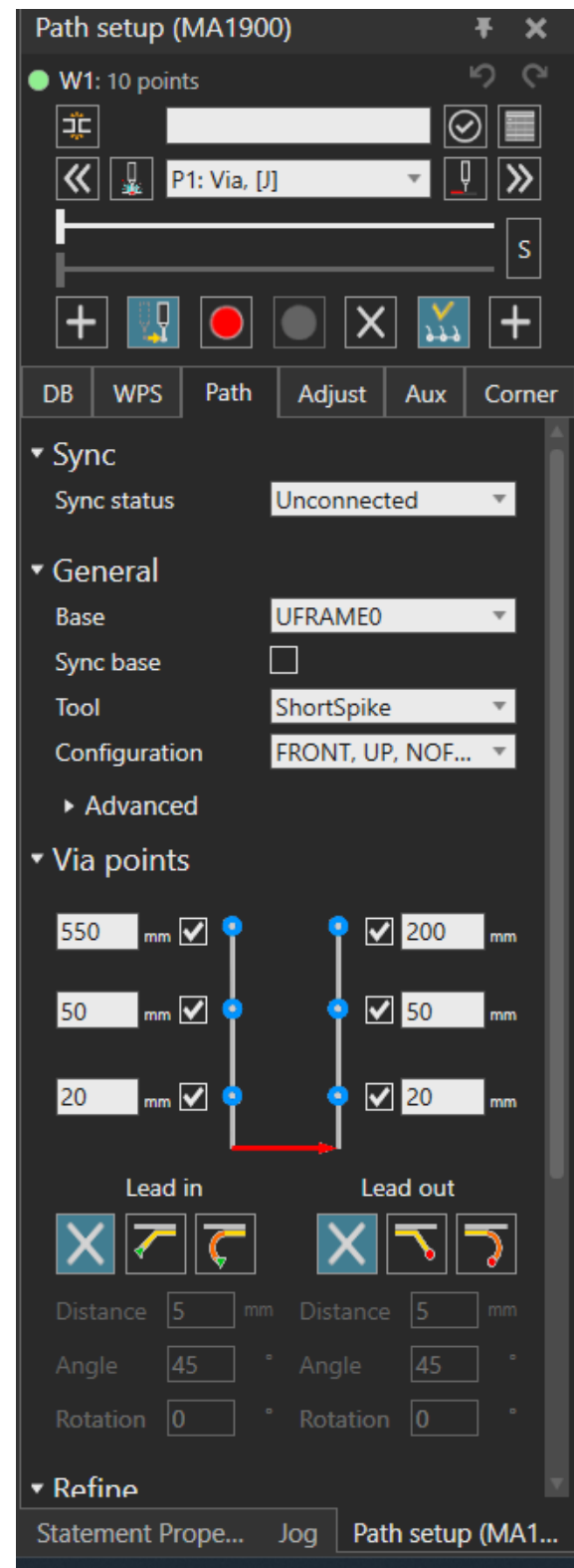


Figure 18: Configuration modifications to adjust collisions

Finally, the overall sequence was run with 'Path Check' to identify any inconsistencies,

collisions or unnecessary tool movements which would cause errors to the weld program and after refinement to the tool movements, green status was achieved which ensured of zero errors in the overall weld program.

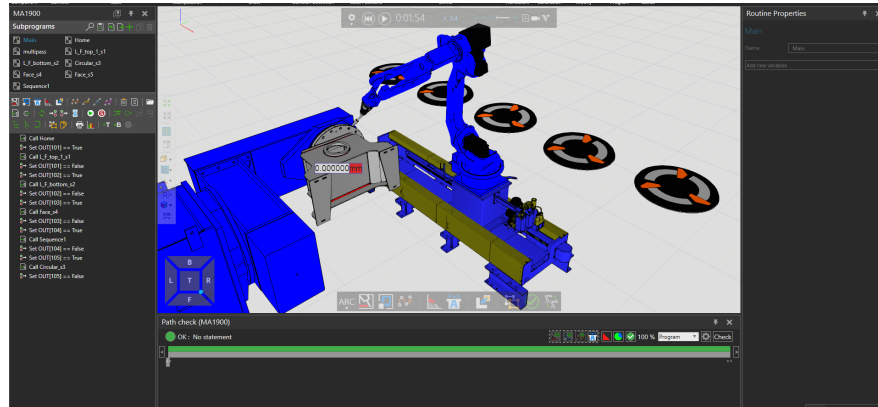


Figure 19: Validation of Path Collision Check

3.2 Statistical Data

3.2.1 Robot Joint Values over Time

In welding sequence analysis, it is important to understand the joint movements which the robot undergoes. This is important to ensure that the joints are not overstrained during movement and that the movements do not affect overall service life of the robot. After initial joint analysis over time, few movements during weld setup were identified to have joint movement leading towards the limit and those movements were adjusted by changing welding sequence and other parameters as angles and offsets. Figure 20 represents the joint movement chart after the final optimization of the weld sequence and tool movement.



Figure 20: Joint Movement measurement

3.2.2 Welding Sequence

Nature of the welds primarily dictated the weld durations in each weld, as the circular, multi pass and the corner welds took much time rather than the simple linear welds. Yet, the sequence was established in such a way that the welds were done in an order with minimal unnecessary movements in workpiece, robot arm as well as the ‘T’ joint of the weld torch. With these considerations taken into account, Figure 21 represents the weld sequence comparison in the optimized version of the weld, with the Multi-pass (Sequence 1), Corner weld (Sequence 2) and the Circular Weld (Sequence 3) taking close to 80% of the total weld duration out of the 5 welds conducted in the program.

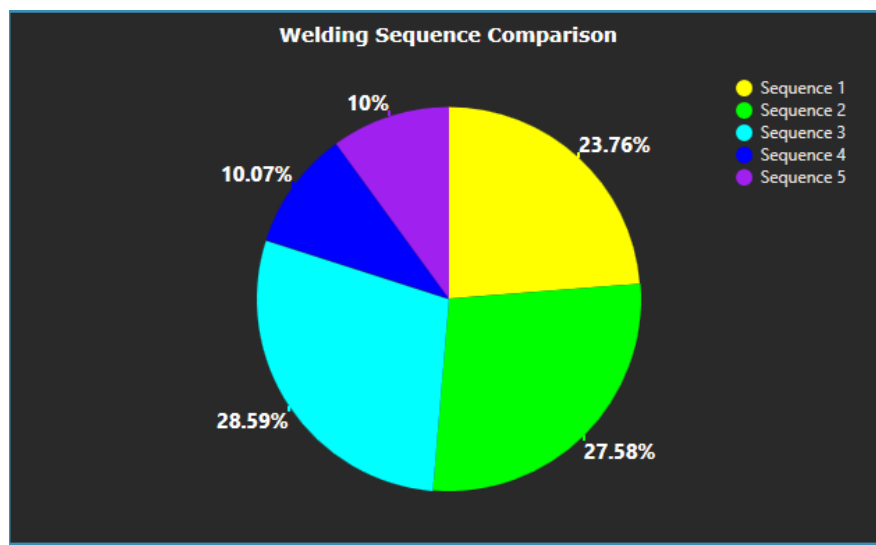


Figure 21: Welding Sequence comparison

3.2.3 Weld Efficiency

Statistical data was used to know the weld duration, weld length, analyses efficiency in individual sequences as well as the overall weld. Further, these data were used to further evaluate the joint movements, and the findings were used to determine the optimal welding sequence with minimal movements. In the final and optimized version of the weld as presented in Figure 22, the overall weld is for 6 minutes and 44 seconds, with a weld length of 3572 mm, with welding occurring on 90.22% out of the total process duration.

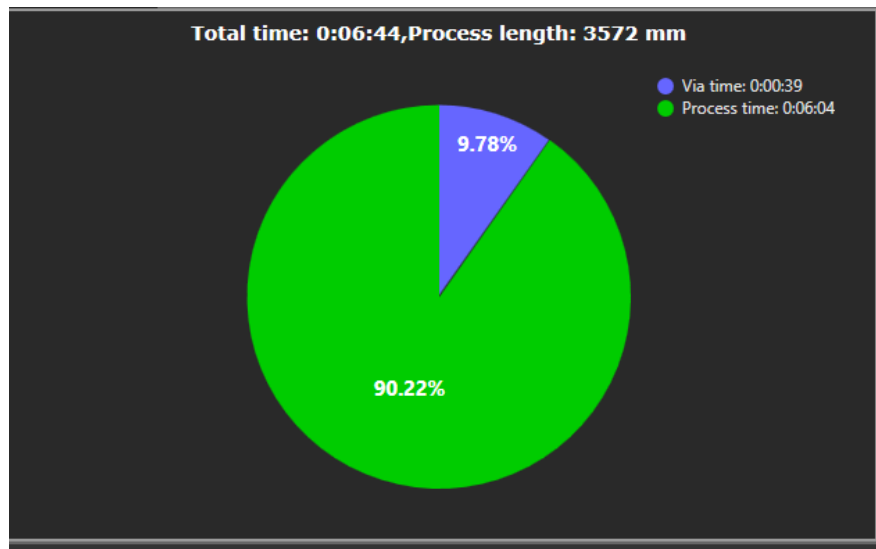


Figure 22: Total Processing Time Comparison

4 Discussion

The welding project covered all the major tasks which need to be carried out during a offline robot programming, from setting up the workstation to optimizing the welding sequence. Though there was a vast array of workstations, positioners, robot and tools to work with, during the built only the compatible selections should be made to ensure that there are no inconsistencies during the later stages of the project. Our initial setup was with an ABB robot (IRB-2600) with a Yaskawa (MA1900) positioner, which was performing well with the linear welds but did not adjust the orientation appropriately in circular welds and corner welds, which was not able to be corrects after multiple attempts. Similarly, combinations from the same manufacture also has to be evaluated beforehand, as there were errors at the final path check which were not detected when running on individual sequence thus making adjustments were not possible. Yaskawa Robot (MA1900) and the positioner, which is presented in Figure 24.

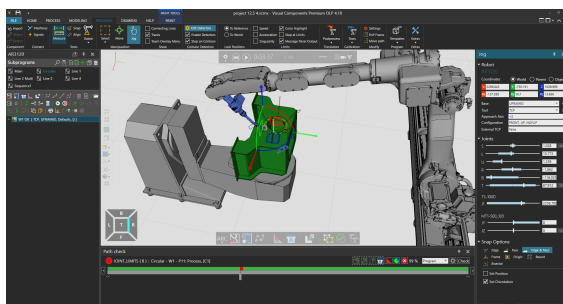


Figure 23: Error in Path Check

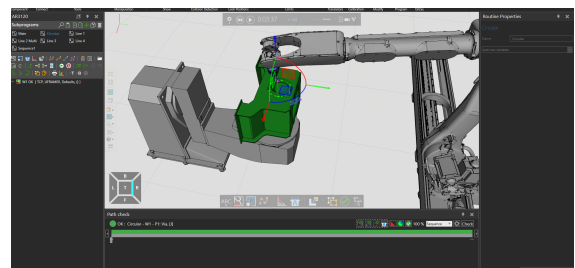


Figure 24: No error shown in the actual sequence

Moreover due to the lack of server space capacity, the initial program that is saved with 0MB and the backup file can only be traced. Therefore, the additional simulation has to be re-setup and ran. However, the process accounts for $\approx 90\%$ in the first file and $\approx 85\%$ in the second file. These compatibility issues caused significant delays in the project, yet with the optimum combinations of positioner and robot, the weld sequence was completed, to achieve the desired results. The optimization of the weld sequences allowed to significant reduction in overall duration and the joint movement analysis led to prevention in unnecessary movements as well as to reduction in unnecessary strain on the systems which lead to longer service life.

5 Conclusion

The project work outlines the importance of the off-line programming methods employed in the industry where significant optimisations can be made. Unlike the online programming methods where the robot is required to be idle during the programming, off-line programming software such as Visual components OLP provides an efficient programming method. Further, the array of possible combinations allows the programmer to choose the ideal combination but the unnecessary errors which develop during the process must be dealt with. Additionally, the realistic visual layout and dimensions that are developed during the setup process allows for the establishment of workstation as per the space availability, prior to any purchases being made.